Office of Research Assurance

Human Subject Research Protections Standard Operating Procedures (SOP)

SOP – 710: Incidental Findings

Number	Effective Date	Version Number	Previous Versions
SOP-710	5/1/2024	v.1	N/A

Purpose

SOP-710 defines the NASA IRB policy and processes with respect to planning for and handling incidental findings during research studies.

An incidental finding is a discovery about an individual that is **unrelated to the goals of the research study**. Incidental findings can either be predictable or unpredictable and can have a medical impact on the participant's health and/or a social behavioral impact not directly related to the participant's medical care. Often, incidental findings are discovered while collecting and analyzing research images and data, but incidental findings may also arise when determining whether a potential research participant qualifies for inclusion in the study population or while collecting baseline physiological information.

Institutional Review Boards (IRBs) have a responsibility to evaluate incidental findings as it directly relates to the Belmont principle of beneficence. Specifically, beneficence provides a directive to avoid harm to participants. To the extent that the return of incidental findings to participants could lead to harm or discomfort, whether physical, psychological, financial, reputational, or social, particularly with no offset of direct benefit, there must be a consideration of whether it is appropriate to return such findings.

Although researchers are not serving in a clinical capacity during research, they have a duty toward research participants that includes identifying, verifying, and communicating incidental findings of clinical significance with regard to the Belmont principle of respect for persons. Additionally, as participants are contributing to the research, there is notion of reciprocity, entitling them to receive information about themselves and have the autonomy to decide to receive information about themselves (Wolf et al., 2008). When the likelihood of identifying incidental findings is high (e.g., whole genome sequencing), it is appropriate to ask the participants in advance if they wish to receive incidental findings. In fact, some genome sequencing studies may also need to consider the return of individual incidental findings to family members of the proband. Legal reporting requirements, such as mandated reporting of communicable diseases, abuses, and intent to harm self or others are an exception.

Federal regulations mandate IRBs evaluate the risks and benefits of research and that those risks and benefits are clearly outlined in the informed consent process. Specifically, the regulations state that informed consent forms include "a statement that significant new findings developed during the course of the research that may relate to the subject's willingness to continue participation will be provided to the subject" (14CFR1230.116(c)(5)). As such, IRBs must consider the risks and benefits of disclosing or returning incidental findings. The disclosure of certain incidental findings might lead participants to pursue lifesaving medical interventions, help participants make informed medical decisions, or identify mission specific risks; whereas, in other scenarios, disclosure could lead to needless further testing, additional incidental findings, costs, anxiety, and distress, with no corresponding medical benefit. Additionally, a lack of disclosure could lead participants to falsely believe they have a clean bill of health (Wolf et al., 2008). As such, it is imperative that researchers explicitly address the possibility of identifying an incidental finding and the risks and benefits associated with the return of information in the protocol and informed consent process. Lastly, in some studies, it may be impossible to provide incidental findings because the participants are de-identified or anonymous (e.g., certain tissue banking activities, research conducted under a waiver of informed consent, exempt studies, and cluster randomized studies).

When determining whether to return incidental findings, the more likely the findings are validated and actionable, the more justification for disclosing (returning) them to participants. Actionable can mean either clinically actionable or personally actionable. Validated usually means the test or diagnosis is conducted using devices or assays that have Food and Drug Administration (FDA) approval or certification under Clinical Laboratory Improvement Amendments of 1988 (CLIA)[1] from the Centers for Medicare and Medicaid Services (CMS). Generally, any incidental finding that reveals a condition or information likely to be life-threatening, a grave condition that can be avoided or ameliorated, or information that affects reproductive decision-making should be reported to participants (Wolf et al., 2008).

	Validated	**Not Validated**
Actionable	Discovered when reviewing results of standard clinical tests (e.g., pregnancy tests, EKGs, MRIs, chest X-rays) • Disclose in a timely manner	Fact dependent and when able, should follow up with validated testing to help provide certainty. • May disclose

[1] Research testing where patient-specific results are reported from the laboratory, and those results will be or could be used 'for the diagnosis, prevention, or treatment of any disease or impairment of, or the assessment of the health of, human beings' is presumed to be subject to CLIA absent evidence to the contrary.

Not Actionable	Fact dependent and may become clinically actionable at a later date • Disclosure recommended	Experimental results (e.g., genetic variant not known to be associated with a disease) • not a strong presumption to disclose

Policy

1. For studies involving NASA astronauts as participants, the Protocol Compliance Officer or the alternate Protocol Compliance Officer, in consultation with the NASA IRB if deemed necessary, will determine if an incidental finding is actionable and/or validated. For studies not involving NASA astronauts as participants, the NASA IRB Chair/designee, in consultation with the NASA IRB or any other experts if deemed necessary, will determine if an incidental finding is actionable and/or validated.
2. For studies involving NASA astronauts as participants, the Protocol Compliance Officer or the alternate Protocol Compliance Officer, in consultation with the NASA IRB if deemed necessary, will determine if an incidental finding reveals a condition or information likely to be life-threatening, a grave condition that can be avoided or ameliorated, or information that affects reproductive decision-making. For studies not involving NASA astronauts as participants, the NASA IRB Chair/designee, in consultation with the NASA IRB or any other experts if deemed necessary, will determine if an incidental finding reveals a condition or information likely to be life-threatening, a grave condition that can be avoided or ameliorated, or information that affects reproductive decision-making.
3. Directly actionable (clinically or otherwise) and validated incidental findings, especially those that reveal a condition or information likely to be life-threatening, a grave condition that can be avoided or ameliorated, or information that affects reproductive decision-making, must be disclosed (or returned) to the research participant in a timely manner unless the participant has elected not to receive such findings.
4. When prospective consent has not been solicited for the return of incidental findings:
    a. For studies involving NASA astronauts as participants, the Protocol Compliance Officer or alternate, in consultation with the NASA IRB if deemed necessary, will determine if actionable but not validated or not actionable but validated incidental findings will be disclosed (or returned) to the participant.
    b. For studies not involving NASA astronauts as participants, the NASA IRB Chair/designee, in consultation the NASA IRB or any other experts if deemed necessary, will determine if actional but not validated or not actionable but validated incidental findings will be disclosed (or returned) to the participant.
5. Incidental findings that are neither actionable nor validated will not be disclosed to participants.

6. When warranted, incidental findings will be returned to participants, or their legally authorized representative, directly (this may be in oral or written form).
7. For studies involving NASA astronauts as participants, the Protocol Compliance or alternate will disclose (return) or is to be involved in the disclosure (return) of an incidental finding. Incidental findings that are determined by the Protocol Compliance Officer or alternate to be a real or potential risk to the mission will require timely notification to the NASA astronaut and his/she/their assigned flight surgeon. After this, incidental findings may be placed in the participant's medical record if necessary.
8. For studies not involving NASA astronauts as participants, an individual with a clinical relationship with the research participant will disclose (return) or is to be involved with the disclosure (return) of an incidental finding. Incidental findings might also be provided to the participant's health care provider or be placed in his/her/their medical record.
9. Principal Investigators (PIs) are not to disclose incidental findings to participants unless otherwise deemed necessary by the NASA IRB.
10. If a researcher has an ethical objection to allowing participants to opt out of receiving clinically significant, actionable, validated, lifesaving incidental findings, he/she/they need not enroll these individuals in the research study and they must outline this exclusion criterion in the protocol and informed consent form.
11. If a researcher discovers a clinically significant, actionable, validated, lifesaving incidental finding for a participant who has opted out of receiving incidental findings, the investigator should seek advice from the NASA IRB. Generally, the PI should reconfirm with the participant who indicated refusal that he/she/they are electing to decline information on all incidental findings, even those with high health importance that can be treated.
12. For studies in which a Reliance Acknowledgment is executed:
    a. When NASA is the IRB of Record, IRB protocols will adhere to this policy
    b. When NASA has ceded review to another institution, the IRB of Record's institutional policies govern, although the NASA IRB reserves the right to make recommendations on a protocol-specific basis

Responsibilities

Principal Investigator

1. Whether or not the researcher is also the treating physician, if the researcher or a member of the research team identifies an incidental finding of potential concern, it is the PI's duty to handle this finding responsibly and promptly.
2. Researchers are encouraged to consult with an expert (e.g., radiologist, clinical geneticist, etc.) to determine which potential incidental findings may be of clinical



significance during protocol development and prior to execution. When an incidental finding is discovered during the course of the research, consultation should not provide information that would disclose the participant's identity to protect the participant's privacy and confidentiality.

3. If a researcher identifies an incidental finding during the course of his/her/their study, the researcher must notify the NASA IRB immediately. This can be done by phone or email, but must be followed up with the following:
    a. Submit reportable new information (RNI) in the eIRB outlining:
        1. the incidental finding;
        2. an opinion as to whether the finding is actionable and/or validated with supporting information;
        3. a recommendation for consultants where appropriate;
        4. an opinion as to whether the finding will be disclosed with justification (including risks and benefits);
        5. if applicable, the disclosure plan; and
        6. if able, confirmation that the participant prospectively agreed to the disclosure
    b. Wait for direction from the NASA IRB regarding recommendations for next steps before taking action

    Note: NASA IRB approval for returning findings may be withheld in the absence of CLIA certifications where appropriate.

4. If there is a known likelihood of identifying incidental findings at the time of IRB submission (whether predictable or unpredictable), researchers must outline such in the protocol and informed consent form.
    a. The protocol must outline:
        1. the probability and types of incidental findings;
        2. a process for handling incidental findings, including consultation to validate the incidental finding (incorporating this policy); and
        3. plans for disclosure with justification (incorporating this policy)
    b. The informed consent form must include:
        1. the potential and likelihood of discovering incidental findings;
        2. examples of potential incidental findings;
        3. the process for handling incidental findings, including consultation to validate the incidental finding (incorporating this policy);
        4. the potential risks/benefits of disclosure;

5. a statement that findings during a research study cannot be assumed to be of clinical quality and therefore, cannot substitute a clinical evaluation; and
6. whether the participants will be informed of the incidental finding(s). If they will be informed, it must be clear that the participant agrees to be contacted and informed of the findings, including the potential risks involved such as anxiety, cost of further clinical evaluation, findings becoming part of their medical record, etc.

Note: Participants should be told if reporting incidental findings will or will not be possible when consenting to data use for future research.

NASA IRB approval for returning findings may be withheld in the absence of CLIA certifications where appropriate.

NASA IRB

1. Upon receipt of an RNI in the eIRB outlining the identification of an incidental finding or findings, the NASA IRB Staff will:
    a. notify the NASA IRB Chair/designee for studies not involving NASA astronauts as participants to determine next steps; and/or
    b. notify the NASA IRB Chair/designee and the Protocol Compliance Officer/alternate for studies involving NASA astronauts as participants to determine next steps
2. The NASA IRB will consider the probability, number of incidental findings, and disclosure plans in proposed study when assessing the risk to benefit ratio and use this information to determine an appropriate level of review.
3. The NASA IRB will evaluate a proposed study's procedures for handling incidental findings as part of their review and approval process.
4. The NASA IRB will provide consultation on incidental findings as appropriate.
5. The NASA IRB will facilitate all communication of incidental findings with the Protocol Compliance Officer when appropriate.

Definitions

1. Incidental finding – An incidental finding is a discovery about an individual that is **unrelated to the goals of the research study**. Incidental findings can either be predictable or unpredictable and can have a medical impact on the participant's health and/or a social behavioral impact not directly related to the participant's medical care
2. Actionable – ethically requiring action - can mean either clinically actionable or personally actionable
3. Validated - usually means the test or diagnosis is conducted using devices or assays that have Food and Drug Administration (FDA) approval or certification

under Clinical Laboratory Improvement Amendments of 1988 (CLIA) from the Centers for Medicare and Medicaid Services (CMS)

4. Protocol Compliance Officer- a physician from NASA SD that is also a NASA IRB member. He/she/they shall have no conflict of interest with potential crewmembers.
5. Alternate Protocol Compliance Officer- an alternate physician from NASA SD, who is a NASA IRB member, that possesses no conflict of interest with potential crewmembers, and fulfills the responsibilities of the protocol compliance officer when he/she/they is/are unable to.

Acronyms

IRB – Institutional Review Board

PI – Principal Investigator

NASA – National Aeronautics and Space Administration

References

1. 14CFR1230
2. Belmont Report
3. Wolf et al. (2008). Managing Incidental Findings in Human Subjects Research. *Journal of Law, Medicine & Ethics, 36*(2), 219-211.
4. Secretary's Advisory Committee on Human Research Protections (SACHRP) Recommendation – Attachment F – Recommendations on Reporting incidental Findings

SOP Role	Name	Date
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